Mederti

Data Asset Inventory

What we hold today — a reference for source mapping

Prepared for: Chief Data Officer

Purpose: Inventory of live, in-production data assets and sources to inform forward source-mapping and prioritisation.

Scope note: All counts are live production figures queried on 3 June 2026. Built-but-unshipped scrapers and pending migrations are excluded from headline numbers and flagged explicitly in Section 9 (Coverage Gaps).

1. Executive Summary

Mederti operates a continuously-refreshed database of global drug-shortage and recall intelligence, enriched with pharmaceutical reference data and a supply-side concentration layer. The platform ingests from official regulators across 20+ jurisdictions on a daily cron, normalises everything to canonical drug identities, and layers causal and macro-signal context on top.

At a glance, the live database holds:

Asset	Live count	What it is
Shortage events	37,065	Structured shortage records (28,418 active) across 20+ countries
Recall records	24,293	Product recalls/withdrawals, 99% linked to a canonical drug
Recall–shortage links	78,897	Causal links between recalls and downstream shortages
Canonical drugs	17,835	Deduplicated molecule/product master
Brand / product catalogue	160,977	Brand & pack-level records rolled up to canonical drugs
API suppliers	14,009	Active-ingredient manufacturer/site signal (FDA DMF + WHO PQ)
Macro signal sources	134	Catalogued early-warning / macro / logistics feeds
Registered data sources	57	Regulator & reference feeds (41 currently active)

Table 1. Headline live data volumes (production, 3 June 2026).

Three structural strengths a CDO should note:

- **Breadth of regulator coverage** — daily ingestion from the major shortage-reporting agencies (FDA, EMA, TGA, MHRA, Health Canada, Swissmedic, PMDA, plus ~15 EU national agencies).
- **Identity resolution** — 160,977 brand/pack records collapse to 17,835 canonical drugs, so a search for a brand rolls up to molecule-level shortage truth.
- **Causal + supply context** — recall→shortage links and a 14k-row API-supplier concentration layer turn raw events into forward signal, not just a feed.

2. Data Domains

The schema (49 migrations) organises into six domains. Counts are live.

Domain	Core tables	Live rows	Status
Shortage core	shortage_events, shortage_status_log, live_status_layer	37,065 events / 56,500 status logs	Mature
Recall core	recalls, recall_shortage_links	24,293 / 78,897 links	Mature
Drug intelligence	drugs, drug_catalogue, drug_synonyms, atc_codes, drug_rxnorm, drug_external_ids	17,835 / 160,977 / 318	Mature
Supply intelligence	api_suppliers, manufacturers, supplier_inventory, supply_intelligence_layer	14,009 / 5 / 28	API layer strong; finished-dose thin
Macro / content	intelligence_sources, intelligence_articles, ai_insights_cache	134 catalogued sources	Catalogue mature; articles building
Ops & audit	data_sources, raw_scrapes, audit_logs	57 / 2,556 / —	Operational

Table 2. Data domains and live volumes.

3. Source Inventory — Regulatory Feeds

Shortage and recall data is sourced exclusively from official national/supranational regulators, ingested daily on a staggered cron (19:00–07:00 UTC). Each source dedupes via an MD5 shortage key, so repeated runs are idempotent. The table lists the regulator feeds live and producing data today.

3.1 Shortage feeds (live, daily)

Region	Agencies live in production
North America	FDA (US), Health Canada (CA)
UK & Ireland	MHRA (UK), HPRA (IE)
EU — supranational	EMA
EU — national	BfArM (DE), ANSM (FR), AIFA (IT), AEMPS (ES), CBG-MEB (NL), DKMA (DK), FIMEA (FI), Läkemedelsverket (SE), SUKL (CZ), OGYEI (HU), Swissmedic (CH), NoMA (NO), AGES (AT)
EU — newer	Greece (EOF), Portugal (INFARMED), Belgium (FAMHP), Poland (MZ)
Asia-Pacific	TGA (AU), HSA (SG), Medsafe + PHARMAC (NZ), PMDA (JP), MFDS (KR), Malaysia (NPRA)
Latin America	ANVISA (BR), COFEPRIS (MX), Argentina (ANMAT)
Africa & Middle East	SAHPRA (ZA), NAFDAC (NG), SFDA (SA), UAE (MOHAP)

Table 3. Regulatory shortage feeds wired into the daily cron.

3.2 Recall feeds (live, daily)

Recall ingestion runs on the same cadence: TGA, FDA (enforcement + MedWatch + Drugs@FDA), Health Canada, EMA, MHRA, AIFA, ANSM, AEMPS, HSA, Medsafe. A dedicated `recall_linker` populates the 78,897 recall→shortage causal links.

4. Geographic Coverage — Shortages

Live shortage-event counts by reporting country. This is the depth dimension a CDO weighs against breadth when prioritising new sources.

Country	Events	Country	Events
Switzerland	5,930	Norway	973
Japan	5,672	Ireland	778
United States	4,469	Germany	740
Italy	4,422	France	575
Canada	3,233	Greece	440
Spain	3,172	Singapore	321
Finland	1,815	New Zealand	290
Australia	1,707	Malaysia	226
Belgium	1,014	United Kingdom	80
Netherlands	1,010	Portugal / UAE / Slovakia	64 / 20 / 13

Table 4. Live shortage events by reporting country (top jurisdictions).

Status mix across all events: **28,418 active**, 6,439 resolved, 2,204 anticipated. **1,589** events are flagged *synthetic* (recall-derived signal, not a primary regulator notice) and are kept distinct so they never inflate the genuine shortage picture.

5. Geographic Coverage — Recalls

Jurisdiction	Recalls	Jurisdiction	Recalls
United States	17,689	United Kingdom	391
Canada	4,597	EU (EMA)	369
Italy	460	New Zealand	186
Australia	390	France	171
		Spain	40

Table 5. Live recall records by jurisdiction. 24,133 of 24,293 (99%) are linked to a canonical drug.

6. Reference & Enrichment Layer

Raw regulator notices are normalised and enriched against authoritative pharmaceutical reference datasets, loaded by 16 importers. This layer is what lets a brand name, a salt form, or an accented INN all resolve to one molecule.

Reference set	Role
WHO ATC / DDD	Therapeutic classification + defined daily dose
RxNorm (NLM)	US clinical drug nomenclature & ingredient mapping
UNII (FDA)	Unique ingredient identifiers for salt-form resolution
SNOMED CT	Clinical terminology cross-walk
EMA EPAR	European authorisation metadata
OECD pharma	Macro pharmaceutical market reference
PharmaCompass	Supplier / API market reference
INN resolution + substance resolver	Salt-stripping & molecule identity (brand→generic rollup)

Table 6. Reference datasets feeding identity resolution and enrichment.

7. Supply-Side Concentration Layer

The `api_suppliers` table (14,009 rows) is Mederti's answer to the single most predictive shortage driver: active-ingredient supply concentration. It is built from the FDA Drug Master File, enriched with DECRS country-of-manufacture data, and badged against the WHO Prequalification list. It replicates and extends the signal behind the Johns Hopkins API-concentration dashboard.

An indicative country distribution (sampled) shows the concentration risk plainly — the active-ingredient base skews heavily to a small number of origins, which is exactly the vulnerability a forward model needs:

- China dominates the sampled API-supplier base, with Canada, Austria, Belgium, Argentina and Brazil forming a long tail.
- Refreshed **quarterly** (Jan/Apr/Jul/Oct), alongside FDA DECRS and WHO PQ.

Finished-dose manufacturer mastering (`manufacturers`, 5 rows) and the supplier marketplace (`supplier_inventory`, 28 rows) are early-stage by comparison — noted as a gap in Section 9.

8. Macro Intelligence Catalogue

Beyond primary regulator data, Mederti maintains a curated catalogue of 134 macro and early-warning sources, classified by signal type. This is the catalogue most directly relevant to forward source-mapping — it already encodes a view of where leading indicators live.

Signal category	Sources	Signal category	Sources
Availability ground-truth	20	Pipeline	7
Logistics	15	Data portals / discovery	7
Macro	12	Sanctions	6
Procurement	11	Trade	5
Early warning	10	Utilization	5
External shocks	9	Corporate disclosure	5
Reference data	8	Public health	4
Pricing	8	Funding & aid flows	2

Table 7. The 134-source macro catalogue by signal category.

Source-priority guidance is encoded throughout: regulators rank above journals, above specialist trade press, above investigative and national press. This tiering is the scaffolding any new source should be slotted into.

9. Coverage Gaps — Whitespace for Source Mapping

The most actionable section for forward planning. These are dimensions where the platform is thin or absent today — candidate priorities for the CDO's source map. All statements reflect live production state.

9.1 Built but not yet producing data

Scraper code exists for these major markets but they return zero live rows — the fastest coverage wins, since the engineering is largely done:

- China (NMPA), India (CDSCO), Israel (MOH), Turkey (TITCK) — high market-size / geopolitical relevance.
- Several recall counterparts (BfArM, HSA, Medsafe recalls) exist as files awaiting wiring.

9.2 Empty or thin data dimensions

Dimension	Live state	Implication
Drug pricing	0 rows	No price/tender signal — a whole forward axis is unpopulated
Demand / utilization	Schema present, sparse	Demand-spike precursors not yet captured
Finished-dose manufacturers	5 rows	Maker signal exists for APIs (14k) but not finished product
Supplier marketplace	28 rows	Two-sided marketplace data is nascent

Table 8. Thin / empty dimensions and what they cost the forward model.

9.3 Operational / trust gaps

- **Source freshness** — last_scraped_at is only partially wired, so there is no per-source freshness signal exposed to users yet.
- **Scraper hosting** — ingestion still runs partly on local cron; a laptop-sleep is a data-gap risk until the Railway migration completes.
- **No public methodology / freshness dashboard** — a credibility lever for institutional citation.

10. Data Lineage & Quality Controls

- **Raw capture** — every scrape is retained in `raw_scrapes` (2,556 rows) for replay/audit.
- **Idempotent ingestion** — MD5 shortage keys mean re-runs never duplicate events.
- **Identity resolution** — 98% of shortage events and 99% of recalls resolve to a canonical drug.
- **Status history** — 56,500 status-log rows preserve the full lifecycle of each shortage (active → anticipated → resolved).
- **Synthetic flagging** — recall-derived signals are tagged so they never contaminate genuine regulator counts.
- **Security** — row-level security enabled; anonymous bulk-read access revoked; immutable audit log on mutations.

11. Recommended Reading Order for Source Mapping

Suggested sequence when overlaying a future-source map on this inventory:

1. Start with Section 9.1 — the built-but-dark markets are the cheapest coverage gains.
2. Then Section 9.2 — pricing and demand are the two missing forward axes most likely to lift predictive power.
3. Cross-reference any candidate against the Section 8 tiering before adding it, to keep source provenance consistent.

All figures in this document can be regenerated on demand from the live database; the query set used is reproducible.